

Demo: PVTR, a Physical–Virtual O-RAN Testbed for FRMCS Slicing

Mohammed Tamimi*, David Kule Mukuhi [†], Leo Mendiboure ^{‡§}, Sylvain Cherrier [†], Rami Langar ^{†¶},
Marion Berbineau [‡]

*DTIPG, SNCF, 93574 La Plaine-St-Denis, France.

[†]LIGM-CNRS UMR-8049, University Gustave Eiffel, 77454 Marne-la-Vallée, France.

[‡]COSYS/ERENA, University Gustave Eiffel, 33400 Talence, France.

[§]LIUPPA, Université de Pau et des Pays de l'Adour (UPPA), 64600 Anglet, France.

[¶]Software Eng. and IT Department, École de Technologie Supérieure (ÉTS), Montréal, QC, Canada.

Abstract—The Future Railway Mobile Communication System (FRMCS) must multiplex safety-critical train control, high-throughput video, and best-effort passenger traffic over bandwidths as narrow as 5–10 MHz. Evaluating dynamic slice control is difficult because simulation does not capture the complete radio and application loop, whereas railway field trials are costly and difficult to reproduce. We present *PVTR*, a hybrid physical–virtual O-RAN testbed that runs the same 5G core, CU/DU, near-real-time RIC, slice configuration, and xApp across two execution modes. Its physical mode closes the control loop over a USRP-based 5G radio and mobile devices. Its virtual mode replaces the over-the-air path with ZeroMQ-connected UEs driven by channel-quality traces collected on a train. As a flagship use case, *RSlice* maps per-slice service state and channel quality to E2SM-RC PRB allocations, with a safety mask protecting critical traffic. The demonstration exposes live telemetry, allocation decisions, and SLA indicators under congestion, and illustrates the complementary experimental conditions provided by over-the-air and railway-trace execution.

Index Terms—FRMCS, O-RAN, network slicing, testbed, radio resource allocation, Railway, xApp.

I. INTRODUCTION

The Future Railway Mobile Communication System (FRMCS) is a 5G-based system designed to support heterogeneous services over bandwidths as narrow as 5–10 MHz [1]. Based on the service classes identified in 3GPP TR 22.889, three railway slices can be considered: a *critical* slice for ultra-reliable, low-latency ETCS, ATO, and emergency voice services; a *performance* slice for medium- to high-rate applications such as CCTV and passenger information systems; and a *business* slice for non-critical services such as onboard Wi-Fi. These slices compete for a finite pool of downlink physical resource blocks (PRBs), the basic schedulable units of radio spectrum.

High-speed mobility produces severe channel fluctuations, making dynamic PRB allocation difficult when safety-critical traffic must remain protected [2]. Most FRMCS scheduling studies are therefore evaluated through simulation or recorded traces: trackside trials require substantial logistics, and representative railway datasets remain scarce. Such evaluations are repeatable and can reproduce high-speed channel evolution, but they do not exercise the complete radio, device, control, and application loop. Conversely, laboratory radio trials expose

implementation and propagation effects, but cannot reproduce train-speed mobility over a railway route.

This demo presents *PVTR*, a hybrid physical–virtual O-RAN testbed that connects these two experimental regimes. *PVTR* implements closed-loop slice control from a near-real-time RAN Intelligent Controller (near-RT RIC) to per-slice PRB enforcement in the srsRAN distributed unit (DU) over the standardized E2 interface using E2SM-RC. The same core, CU/DU, RIC, slice configuration, traffic, and xApp operate in two interchangeable execution modes. The physical mode uses a USRP-based radio, two mobile unmanned ground vehicles (UGVs), and a smartphone. The virtual mode replaces the over-the-air path with ZeroMQ (ZMQ)-connected UEs driven by channel quality indicator (CQI) traces collected on an operational train [3].

PVTR main contribution is an integrated environment evaluating an unchanged O-RAN controller under live and repeatable trace-driven railway conditions while preserving the full network control stack. We demonstrate the platform with *RSlice*, a learning-based xApp trained on railway data, alongside conventional and learning-based schedulers [4]. *RSlice* serves as a demanding use case for demonstrating enforceable, slice-aware control under congestion; the demo does not seek to establish a new algorithmic ranking among the evaluated policies.

II. RELATED WORK

Open-source 5G stacks and near-RT RICs enable programmable RAN experiments, but not directly a reproducible closed loop for slice-aware control. *PVTR* connects the E2SM-RC path to the DU scheduler, enforces per-slice PRB allocation, and consistently orchestrates the RIC, RAN, core, traffic, and monitoring functions across physical and trace-driven execution.

Prior FRMCS slice control [2] and generic RAN slicing schedulers [4] have primarily been evaluated in simulation or against recorded channel conditions. *PVTR* complements these studies by executing such controllers on a physical O-RAN stack and by replaying train-collected CQI traces through the same control and application environment. Its novelty therefore lies in bridging repeatable railway-channel

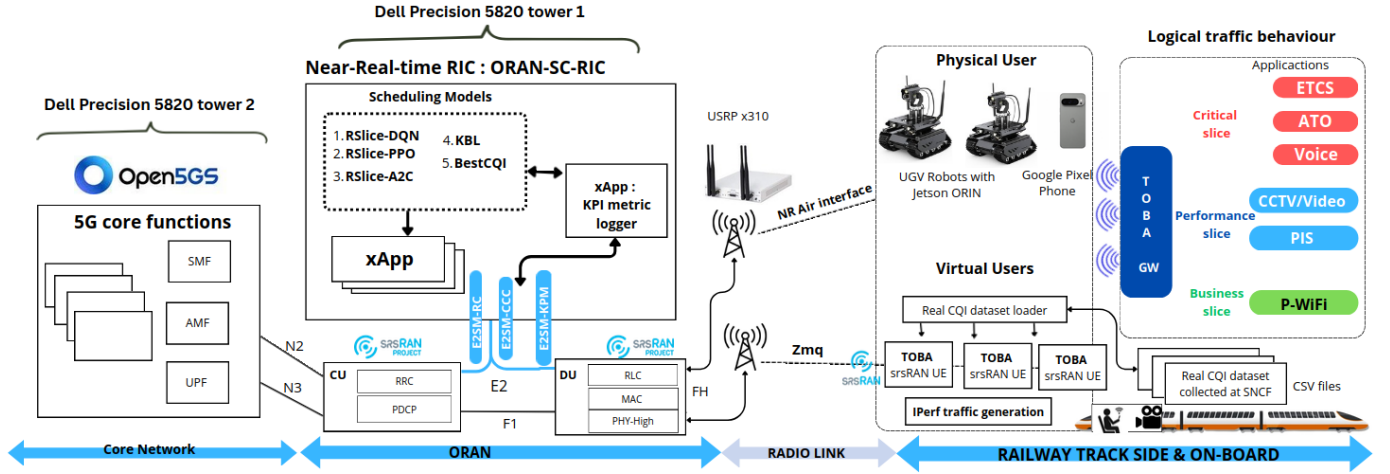


Fig. 1. Architecture of our Physical-Virtual Testbed for Railways FRMCS

evaluation and over-the-air closed-loop validation, rather than in introducing another isolated scheduling algorithm.

III. DESIGN AND IMPLEMENTATION

As shown in Fig. 1, PVTR realizes a disaggregated 5G system in which a near-RT RIC xApp controls per-slice radio resources in a closed loop. The platform comprises three main blocks:

5G core and disaggregated RAN. The RAN follows the O-RAN functional split into a central unit (CU), a distributed unit (DU), and a radio unit (RU), implemented with the srsRAN Project. An Open5GS core provides access and mobility management, session management, and user-plane functions. Three slices are configured end-to-end from core subscriber profiles down to per-slice radio scheduling, and each UE retains the same slice mapping in both execution modes.

RIC and RSlice xApp. The control application runs on the O-RAN Software Community near-RT RIC. At each decision interval, it retrieves per-UE CQI and delivered downlink rate through the platform monitoring interface, aggregates these observations into a per-slice state, and computes the next allocation. The resulting minimum PRB share for each slice is then sent to the DU scheduler through E2SM-RC, closing the observe-decide-enforce loop. RSlice maps the current per-slice service satisfaction and channel quality to a PRB allocation, while a rule-based safety mask excludes actions that could starve the critical slice. The agents are trained on channel traces collected from trains operating at speeds of up to 350 km/h [2]. The platform provides interchangeable Deep reinforcement learning agents (RSlice-PPO and RSlice-A2C), a low-inference-cost value-based agent suited for near-RT operation (RSlice-DQN), the well known channel-only best-CQI scheduler, and the kernel-based KBL method [4].

Hybrid railway environment. The distinguishing feature of PVTR is that the same core, CU/DU, RIC, controller, slice configuration, and traffic run across two execution modes. In *physical mode*, the DU drives a software-defined radio and real

UEs communicate over the air. In *virtual mode*, the over-the-air path is replaced by a ZeroMQ (ZMQ) transport and UEs replay train-collected CQI measurements. This mode supports repeatable policy comparisons, controlled traffic and load variations, and the reproduction of channel evolution along railway routes. Switching modes changes radio execution conditions without changing the evaluated control logic.

IV. DEMO SETUP

The platform is deployed on two Dell Precision workstations, as illustrated in Fig. 2. *Tower 1* hosts the srsRAN CU/DU, a USRP X310 equipped with two UBX-160 daughterboards and connected over 10 GbE, the containerized near-RT RIC, the RSlice xApp, and a metrics logger. *Tower 2* hosts the Open5GS core and the downlink traffic generator, connected to Tower 1 through standard 5G N2 and N3 interfaces.

The radio operates in a 5G New Radio (NR) FDD configuration with 15 kHz subcarrier spacing within the narrow FRMCS spectrum. Three physical UEs attach over the air and are mapped to distinct slices, allowing the scheduler's decisions to be observed simultaneously through radio measurements, application throughput, and service-level indicators. Two mobile UGVs, each equipped with a Jetson Orin Nano and a Quectel RM530N-GL modem, represent critical and performance services. A Google Pixel smartphone carries business-slice traffic. The PVTR configuration and RSlice implementations are publicly available to support reproducibility [5].

V. DEMONSTRATION PROCEDURE AND RESULTS

The demonstration reproduces a congested FRMCS cell in which services of unequal priority compete for radio resources. The critical slice carries a low-rate, delay-sensitive stream representing train-control signaling; the performance slice carries a medium-rate stream representing video surveillance; and the business slice carries best-effort video representing passenger connectivity. Their aggregate offered load exceeds

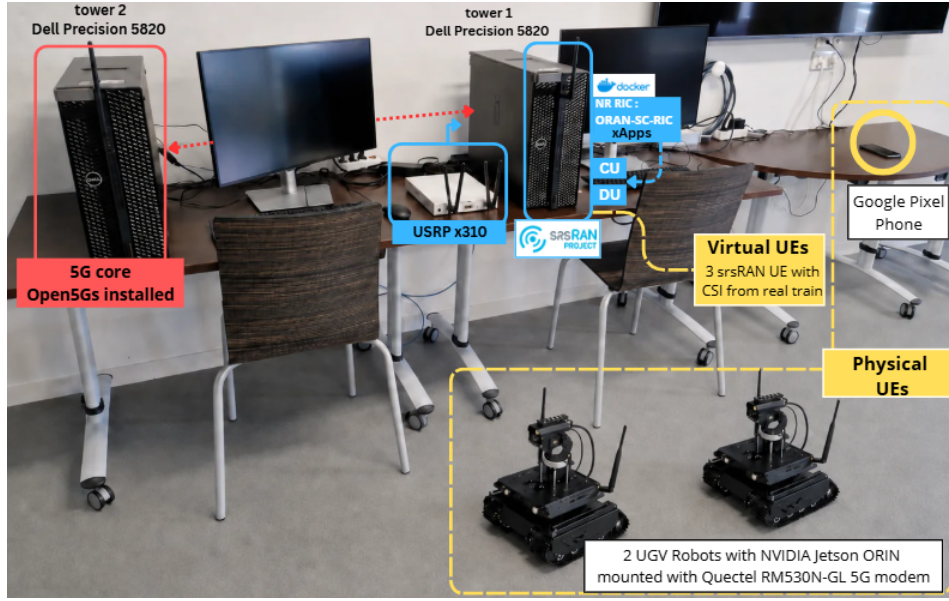


Fig. 2. Our PVTR Testbed for Railways FRMCS

the cell capacity, so the selected scheduler directly determines which service targets are met.

For each run, the operator selects an execution mode and one of the available scheduling policies. The same traffic scenario can then be repeated with another controller or execution mode while retaining the service definitions and slice mapping. The dashboard displays CQI, delivered down-link rates, PRB decisions, and slice-specific SLA indicators computed against the configured service targets. In physical mode, UGV movement induces live channel variations. In virtual mode, the platform replays a selected railway trace, including channel conditions collected at speeds of up to 350 km/h. The controller reacts at each decision interval, making the effect of policy changes visible without modifying the slice or traffic configuration.

Experiment 1: closed-loop allocation on physical hardware. Fig. 3 reports per-slice SLA satisfaction during a representative overloaded physical-mode run. The RSlice agents maintain critical-slice satisfaction at approximately 0.9–1.0 and assign the residual capacity according to service priority. KBL produces a similar priority-oriented profile. In contrast, best-CQI improves satisfaction for the performance and business UEs but provides weaker protection to the critical slice because it allocates resources according to instantaneous channel quality rather than service criticality. The experiment demonstrates that the selected xApps produce distinct, enforceable PRB decisions on the physical RAN; it is not intended as a statistically significant ranking of the scheduling algorithms.

Experiment 2: complementary physical and virtual regimes. Fig. 4 reports the SLA violations observed when the same controller is executed over the physical radio and the ZMQ trace-replay path. The physical run shows stronger protection of the critical slice and larger violations for lower-

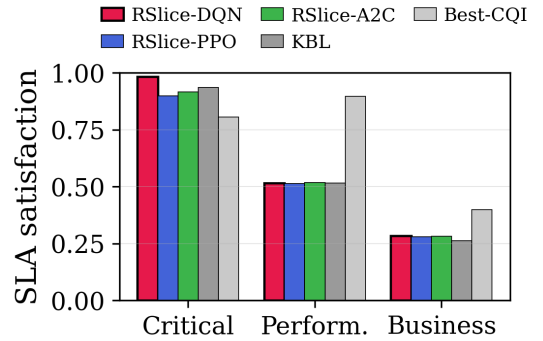


Fig. 3. Slice SLA satisfaction in overloaded physical-mode

priority traffic, whereas the trace-driven run exposes a different service profile under high-speed railway channel evolution. Because the two runs do not follow matched channel trajectories, their values should not be interpreted as a controlled performance comparison between physical and virtual execution. Instead, they illustrate the complementarity of the two modes: physical execution captures real radio, device, and short-range mobility effects, while virtual execution provides repeatable access to channel dynamics that the UGVs cannot reproduce, including traces collected at up to 350 km/h.

VI. CONCLUSION

We presented PVTR, a hybrid physical-virtual O-RAN testbed that executes the same slice controller and service configuration across a USRP-based radio path and a railway-trace replay path. RSlice demonstrates closed-loop per-slice PRB control through E2SM-RC and enforces the intended service priority under congestion. The two PVTR modes provide complementary capabilities: repeatable evaluation under

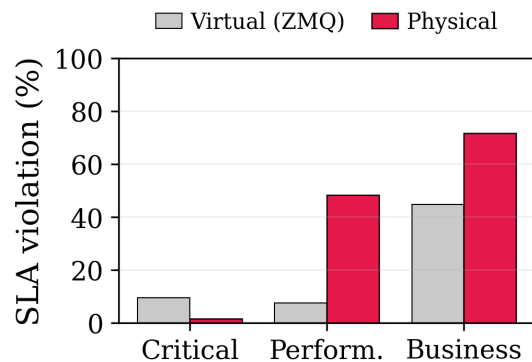


Fig. 4. SLA violations in physical-radio vs railway-trace replay Virtual mode.

channel dynamics collected from high-speed trains and live execution involving real radios, devices, traffic, and mobility. Their different outcomes also show why trace-driven results should not be assumed to transfer unchanged to physical deployments. While the current demonstration focuses on single-cell downlink slicing rather than full railway mobility management, the released control and measurement substrate can support further railway xApps for handover, admission control, and energy-aware RAN operation.

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